

Cybersecurity Incident Report

Scenario:

You are a cybersecurity analyst working at a company that specializes in providing IT services for clients. Several customers of clients reported that they were not able to access the client company website www.yummyrecipesforme.com, and saw the error “destination port unreachable” after waiting for the page to load.

You are tasked with analyzing the situation and determining which network protocol was affected during this incident. To start, you attempt to visit the website and you also receive the error “destination port unreachable.” To troubleshoot the issue, you load your network analyzer tool, tcpdump, and attempt to load the webpage again. To load the webpage, your browser sends a query to a DNS server via the UDP protocol to retrieve the IP address for the website's domain name; this is part of the DNS protocol. Your browser then uses this IP address as the destination IP for sending an HTTPS request to the web server to display the webpage. The analyzer shows that when you send UDP packets to the DNS server, you receive ICMP packets containing the error message: “udp port 53 unreachable.”

```
13:24:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?
yummyrecipesforme.com. (24)
```

```
13:24:36.098564 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 254
```

```
13:26:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?
yummyrecipesforme.com. (24)
```

```
13:27:15.934126 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 320
```

```
13:28:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?
yummyrecipesforme.com. (24)
```

```
13:28:50.022967 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2
udp port 53 unreachable length 150
```

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic

log

The network protocol analyser logs indicate that port 443 is unattainable when trying to access the secure employee background check website. Port 443 is normally used for HTTPS traffic. This may indicate a problem with the web server or the firewall configuration. It is possible that this is an indication of a malicious attack on the web server.

Part 2: Explain your analysis of the data and provide at least one cause of the incident

The incident happened in the morning when the (HR) team reported that they could not reach the background check web portal. The network security team responded and began running tests with the network protocol analyzer tool

. The resulting logs revealed that port 443, which is used for HTTPS traffic, is not accessible . We are continuing to investigate the root cause of the issue to determine how we can restore access to the secure web portal. The next steps include checking the firewall configuration to see if port 443 is blocked and contacting the system administrator for the web server to have them check the system for signs of an attack. The HR team believes it is possible that a certain new hire may want to keep them from performing the background check. The network security team suspects this person might have launched an attack to crash the background check website. After examining I find out it was a cyber attack as the wrong port number tried gaining access